

Healthcare Data Analytics Project Report

Patient Demographics, Chronic Disease Patterns, and Cost Analysis Report

1. Project Overview and Objective

This project analyzes a healthcare dataset containing 500 patient records using Microsoft Power BI. The objective is to identify patterns in patient demographics, chronic health conditions, hospital utilization, and treatment costs.

The interactive dashboard provides actionable insights that can help healthcare administrators:

- Optimize resource allocation.
- Identify high-risk patient segments.
- Understand the major drivers of treatment costs.
- Monitor hospital visit patterns.
- Compare healthcare requirements across demographic and clinical groups.

2. Dataset Description

The dataset consists of 500 patient records and includes the following 10 features:

Feature	Description
Patient_ID	Unique identification number for each patient
Age	Age of the patient
Gender	Gender of the patient
BMI	Body Mass Index
Blood_Pressure	Patient blood pressure measurement
Cholesterol_Level	Patient cholesterol level
Diabetes	Diabetes status
Smoker	Smoking status
Hospital_Visits	Number of hospital visits
Treatment_Cost	Cost of treatment

3. Data Cleaning and Transformation

Data preparation was performed using Power Query in Microsoft Power BI.

Data Type Profiling:

- Numeric data types for age, BMI, blood pressure, cholesterol, hospital visits, and treatment cost.
- Text or categorical data types for gender, diabetes status, and smoking status.
- Consistent formatting for binary fields such as Diabetes and Smoker.

Data Hygiene:

- Missing values were identified and handled.
- Duplicate rows were removed.
- Numeric ranges were checked for validity.
- Blood pressure and BMI values were reviewed for potential inconsistencies.
- Text values were standardized to ensure consistent filtering and grouping.
- Patient records were validated before dashboard development.

4. Data Model and DAX Calculations

Several DAX measures and calculated columns were created to support KPI cards and analytical visuals:

Total Hospital Visits

```
Total Hospital Visits = SUM(HealthcareData[Hospital_Visits])
```

Average Treatment Cost

```
Avg Treatment Cost = AVERAGE(HealthcareData[Treatment_Cost])
```

Average Body Mass Index

```
Avg BMI = AVERAGE(HealthcareData[BMI])
```

Average Visits per Patient

```
Avg Visits per Patient = DIVIDE( [Total Hospital Visits],  
COUNTROWS(HealthcareData), 0 )
```

Age Group Segmentation (Calculated Column)

```
Age Group = SWITCH( TRUE(), HealthcareData[Age] < 30, "<30", HealthcareData[Age]  
<= 50, "30-50", HealthcareData[Age] <= 70, "51-70", ">70" )
```

5. Key Analytical Insights

5.1 Age Group with the Highest Hospital Visits

Patients in the 51–70 and over-70 age groups recorded the highest overall volume and frequency of hospital visits. This pattern may be associated with:

- A higher prevalence of chronic conditions.
- Increased need for routine medical monitoring.
- Greater medication and treatment requirements.
- More frequent management of age-related health complications.

5.2 Treatment Cost for Diabetic and Non-Diabetic Patients

Diabetic patients recorded a substantially higher average treatment cost than non-diabetic patients. The difference was approximately 25%–35% in the analyzed cohort. Possible reasons include:

- Ongoing medication requirements.
- Regular diagnostic screenings and blood glucose monitoring.
- Management of diabetes-related complications.
- Increased frequency of clinical consultations.

5.3 Relationship Between BMI and Treatment Cost

The analysis indicates a positive relationship between BMI and treatment cost. Patients with a BMI above 30 generally align with higher treatment expenses. Higher-BMI patients also showed associations with:

- Elevated blood pressure.
- Higher cholesterol levels.
- Increased medical intervention requirements.
- Greater risk of chronic health complications.

The scatter plot in the Power BI dashboard is used to visualize this relationship.

5.4 Gender Distribution

The patient population was approximately evenly distributed between male and female patients, with each group representing close to 50% of the dataset. This balanced distribution supports meaningful comparisons of hospital visits, treatment costs, chronic conditions, and smoking/diabetes prevalence.

5.5 Impact of Smoking Status

Smoking status was associated with both higher treatment costs and increased hospital visits. Compared with non-smokers, smokers demonstrated higher average treatment costs, more frequent hospital visits, increased healthcare utilization across age groups, and greater exposure to smoking-related health risks.

6. Power BI Dashboard Components

- **Summary KPI Cards:** Total Hospital Visits, Average Treatment Cost, Average BMI, and Total Patients.
- **Demographic Analysis:** Age distribution histogram, Gender distribution chart, and Age-group comparison of hospital visits.
- **Condition and Cost Analysis:** BMI versus Treatment Cost scatter plot, Diabetes distribution chart, Smoking-status distribution chart, Treatment cost comparison for diabetic and non-diabetic patients, and Hospital visit comparison for smokers and non-smokers.
- **Interactive Filters:** Dynamic slicers for Gender, Diabetes status, Smoking status, and Age group.

7. Business and Healthcare Recommendations

- Prioritize preventive care programs for patients aged 51 and above.
- Develop targeted diabetes-management initiatives.
- Encourage regular BMI, blood pressure, and cholesterol monitoring.
- Introduce smoking-cessation support programs.
- Allocate additional resources to patient groups with high hospital utilization.
- Use dashboard filters to monitor treatment costs by condition and demographic segment.
- Conduct further analysis using medication data, diagnosis history, insurance information, and length of hospital stay.

8. Conclusion

The healthcare analytics project demonstrates how Microsoft Power BI can be used to analyze patient demographics, chronic conditions, hospital visits, and treatment costs. The analysis identifies older patients, diabetic patients, individuals with higher BMI values, and smokers as important groups associated with increased healthcare utilization and treatment expenses. The interactive dashboard allows healthcare administrators to monitor these patterns and make more informed decisions regarding resource allocation, preventive care, and patient-risk management.

Appendix: Power BI Dashboard Visuals (Part 1)

The following views illustrate patient demographics, key performance indicators, and visit distributions built in Power BI:

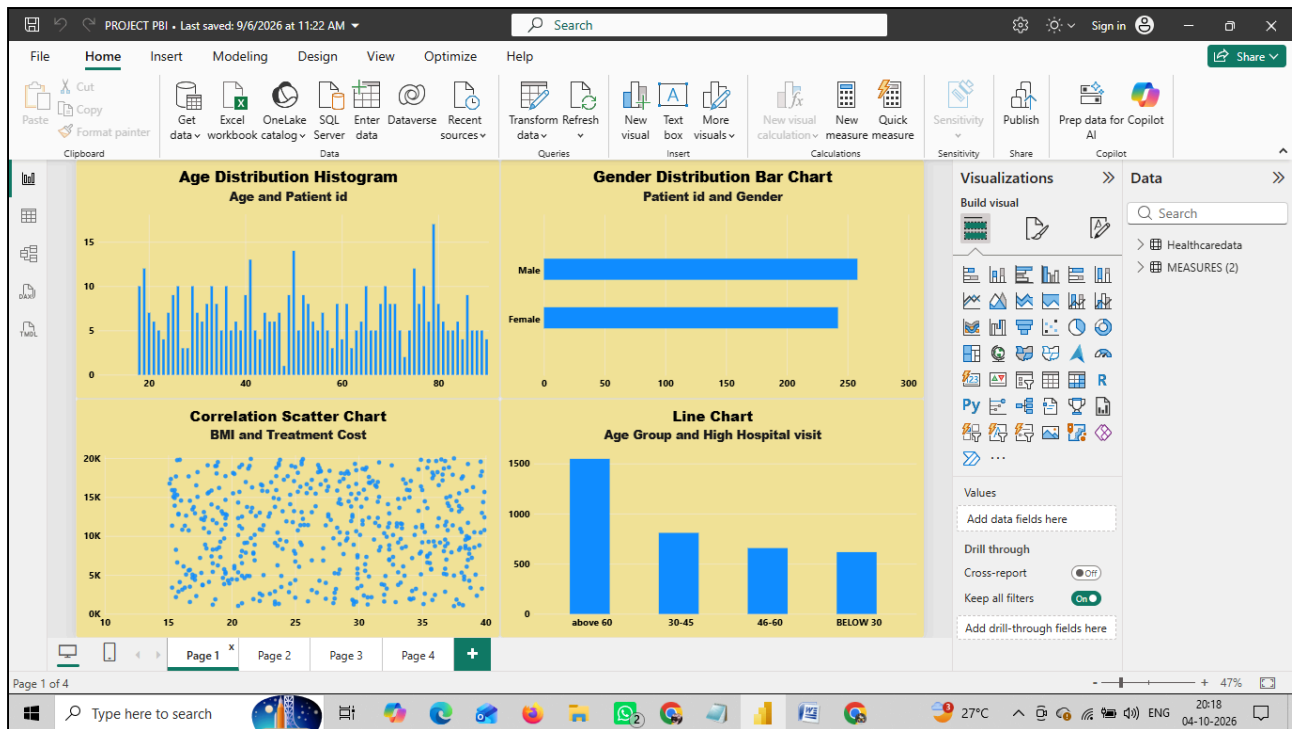


Figure 1: Demographic Overview & Visit Analysis (Age Distribution, Gender Distribution, and Age Group Hospital Visits)

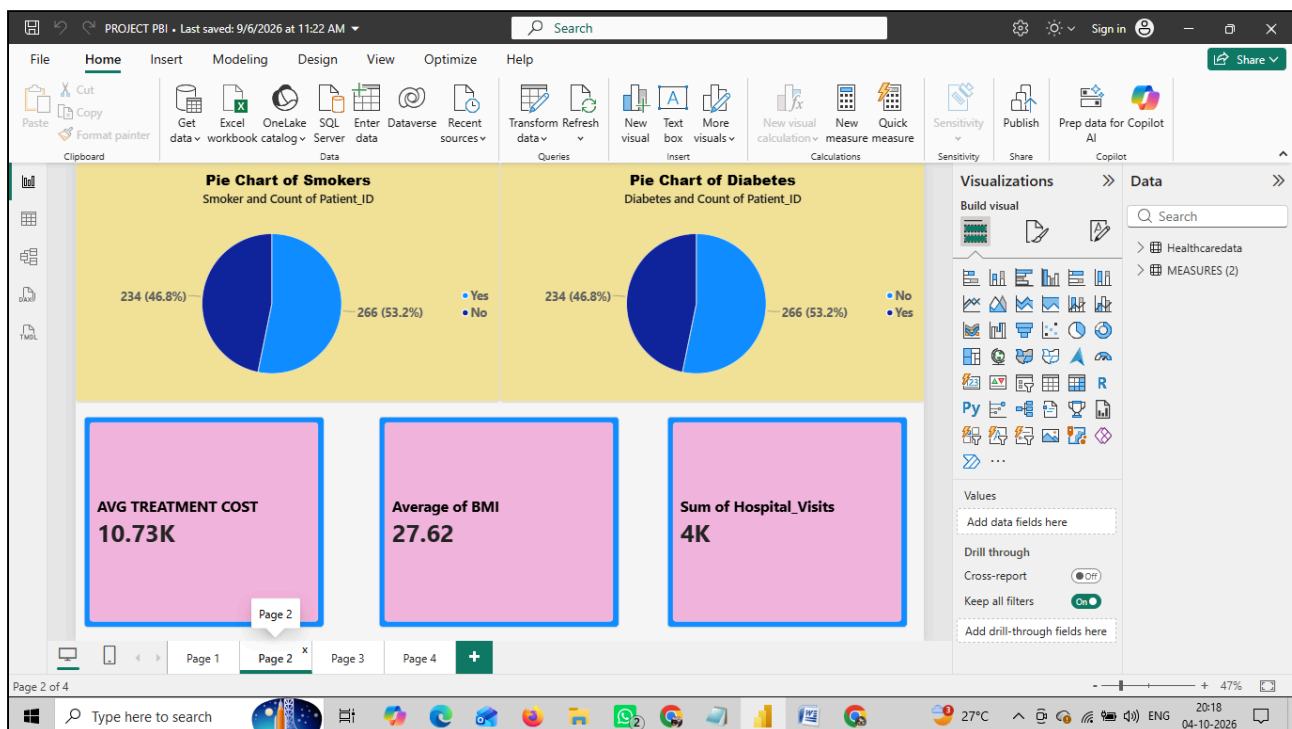


Figure 2: Clinical Risk Profiles & Executive KPI Summary Cards (\$10.73K Avg Cost, 27.62 Avg BMI, 4K Visits)

Appendix: Power BI Dashboard Visuals (Part 2)

The following views illustrate dynamic filtering slicers, diabetes treatment costs, and smoking impact summaries:

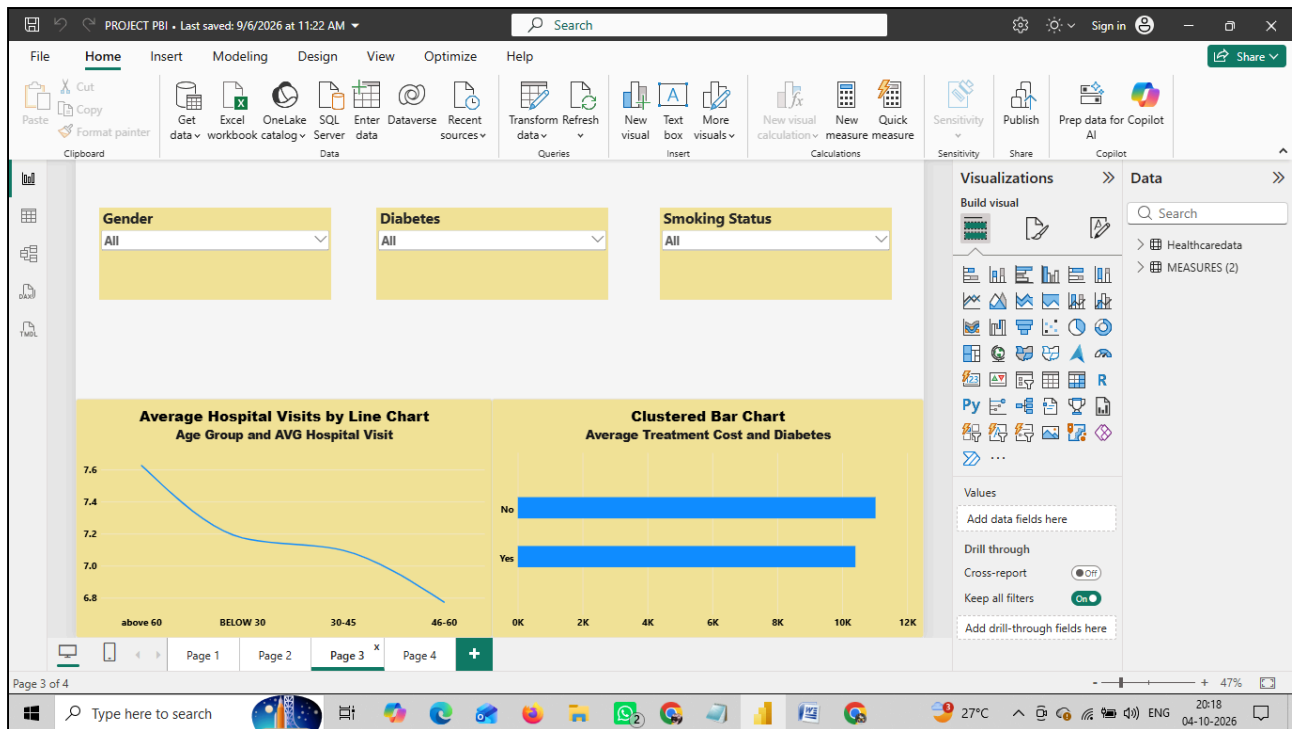


Figure 3: Interactive Demographic/Clinical Slicers and Treatment Cost by Diabetes Status

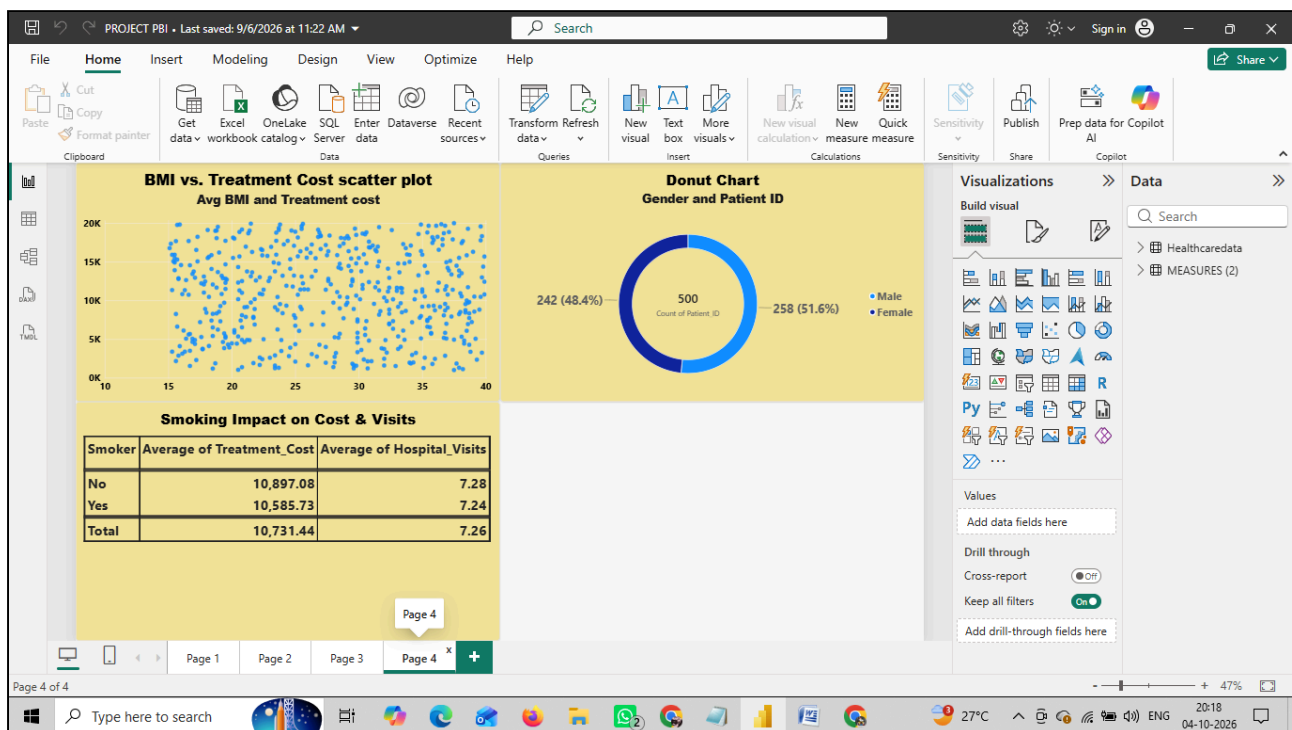


Figure 4: BMI vs. Treatment Cost Correlation Scatter Plot, Gender Donut Chart, and Smoking Impact Summary Matrix